Venkatesh Iyer

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Summary

I am a machine learning engineer with 6+ years of experience and proficiency spanning across areas of aviation, medical imaging, and autonomous driving. I have experience in exploratory data analysis, data visualization, developing and deploying machine learning applications in corporate and academic environments.

Technologies

Frameworks/ Libraries: Pytorch, Tensorflow, LangChain, LangGraph, Scikit-learn, OpenCV

Languages: Python, JavaScript (Basic)

Deployment: Flask, Streamlit, Docker, Github Actions (CI/ CD)

Cloud: AWS, Azure (Basic), GCP (Vertex AI)

Database/ Vector Database: MySQL, Chroma, Milvus

Version control: Git

Scrum: Jira

Experience

Sanofi, Data Scientist II, Hyderabad, India **Nov' 25 - Present**

- Developing a Deep Learning pipeline for automated detection, segmentation, and classification of lung nodules in CT imagery. Implementing advanced Computer Vision architectures to enhance diagnostic precision and support early-stage oncology screening
- Developing a large-scale Agentic RAG application integrating 30+ internal and external data sources, including scientific journals and web-based repositories, to automate R&D literature searches

Eli Lilly and Company, AI Engineer II, Bengaluru, India **Oct '24 - Nov' 25**

- Developed a legal application utilizing Azure AI services and LLM prompting to extract and highlight structured legal data and generate case summaries from lawsuit documents.

Tarento Technologies, Senior AI Engineer, Bengaluru, India **April '22 - Sept '24**

- Developed a conversational interface using a document-based chatbot (RAG) by leveraging the Azure-hosted LLM capability to answer queries from a collection of documents
- Developed RAG POCs as a comparison study between different embedding models (Hugging Face and Open AI), LLMs (GPT 3.5, GPT 4), vector databases (FAISS, Milvus), and its index creation and search algorithms to understand the performance in terms of accuracy and speed of response
- Led an edge-based OCR open source solution (SARAL) to convert physical data to digital data. As a part of this, I developed and fine-tuned the digit and alpha-numeric models along with the detection of OMR circles using traditional computer vision techniques. This application has been used to assess the mark sheets of 1.75+ crore students in the state of Uttar Pradesh
- Mentored a team of 3 in building an open-source tool for an initiative (C4GT) to create layout designs. This tool considerably reduced the number of man-hours to generate layouts
- Developed a pipeline to digitize the electricity and gas consumption invoice bills using OCR Deep learning techniques

German Aerospace Center (DLR), Machine Learning Researcher, Germany **Oct '20 - Jul '21**

- Research on the feasibility of the cockpit-based flight assistance system that detects future anomalies using time-stamped data such as cruising speed, altitude, air space violations, etc., and raising the alert to the pilot for safe flight maneuver
- Developed a univariate time series forecasting POC based on LSTM architecture using time-stamped data extracted from the flight simulator to analyze the viability

Ernst and Young (EY), Associate Software Developer **Jul '14 - Aug '16**

- Led a team of 4 for one of the leading investment banks to manage internal risk audit (Sarbanes-Oxley Act) data using automated workflows on Microsoft SharePoint
- Developed an automated tool to export large amounts of SharePoint list items to MS Excel. This tool was capable of processing data above the default threshold limit
- Designed, developed, and supported live applications and POCs using C#

Education

Technical University Kaiserslautern, M.Sc. in Computer Science (Machine Learning) **Oct '16 - Oct '20**

- **Coursework:** Machine Learning, Deep Learning, Applications of Artificial Intelligence, Computer Vision, Requirements Engineering, Psychology

Gujarat Technological University, B.E in Computer Science **Jun '10 - Jun '14**

- **Coursework:** Operating Systems, Data Structure and Algorithms, Compilers, Computer Graphics

Projects

Toxic (Obscene) comment classification **Jun '20 - Jul '20**

- Developed a web application using a pre-trained Hugging Face transformer model that classifies the toxicity of comments on a YouTube video

Semantic segmentation and object detection using a common pipeline **Jun '19 - Jan '20**

- Researched methodologies to assess the performance of both semantic segmentation and object detection when performed in parallel using a one-step network architecture
- Designed and implemented a combined multi-task CNN architecture using a common encoder to extract features and separate decoders for both tasks. This approach reduced the training time and the number of trainable parameters.

Cancer Nuclei Segmentation **Feb '20 - Apr '20**

- Implemented the UNet architecture to understand the working of medical image semantic segmentation using the Science Bowl competition dataset.

Awards

Tarento Team Impact Award (Q2-2022)

Tarento Team of the Year Award (Q2-2023)

Tarento Spot Award (Q1-2024, Q2-2023)

Additional information

Languages: English (fluent), Hindi (fluent), Tamil (mother-tongue)

Hobbies: Reading, traveling, cooking

I hereby declare that all the information given above is true and correct to the best of my knowledge